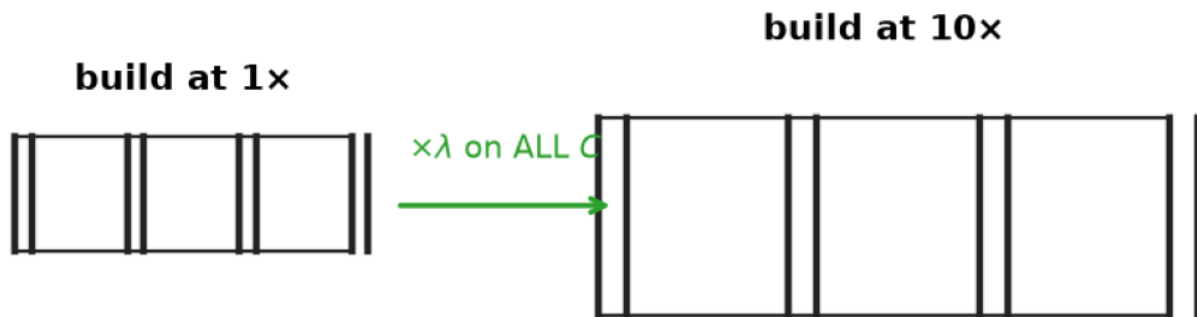


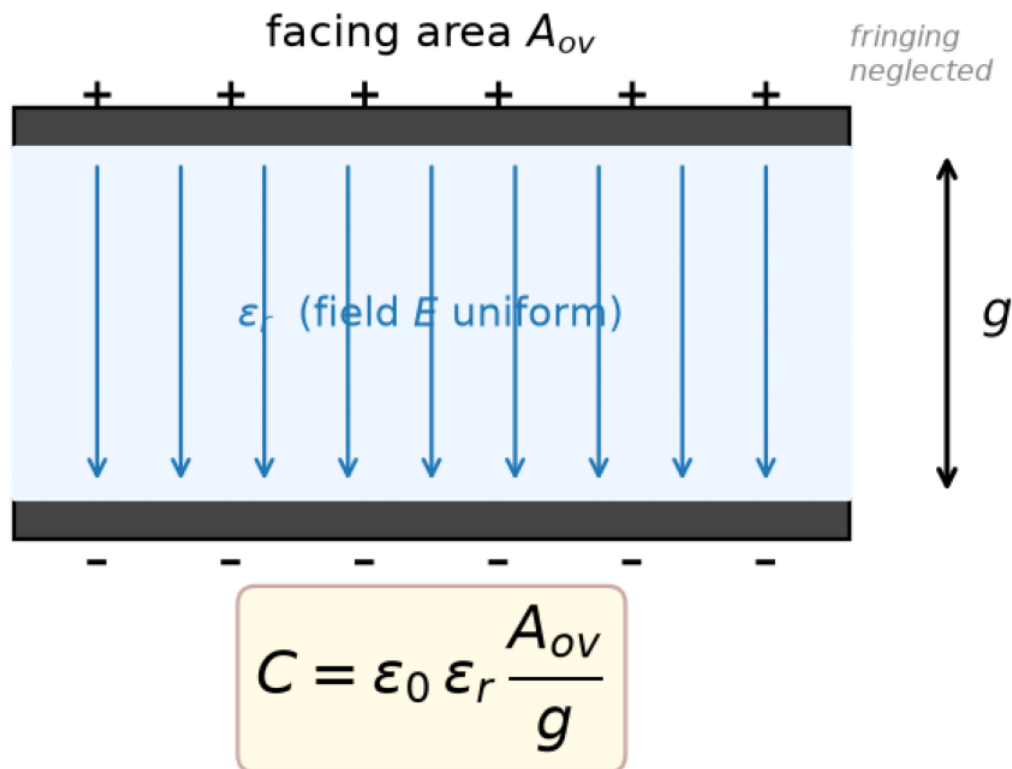
E1 — Scale-free pump gain (Eq. 1)



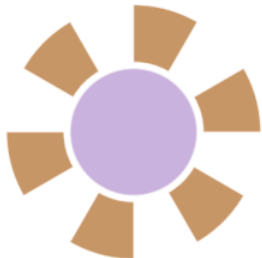
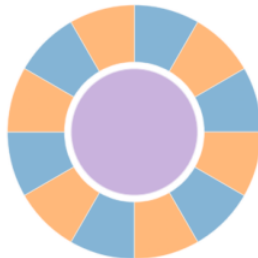
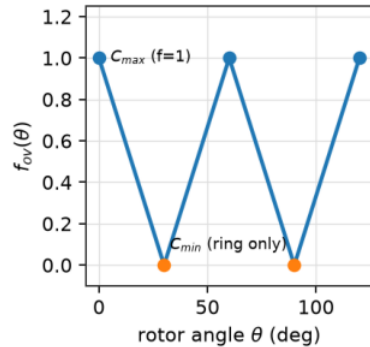
$$Z = Z(K_C, C_a/C_{max}, C_{par}/C_{min}) = 1.334$$

same z at any absolute size — scale ONE family alone $\Rightarrow z$ collapses $\times$

E2 — Parallel-plate base law (Eq. 2)



E3 — Sector overlap, the two extremes, swing ratio (Eqs. 3, 5, 6, 7)

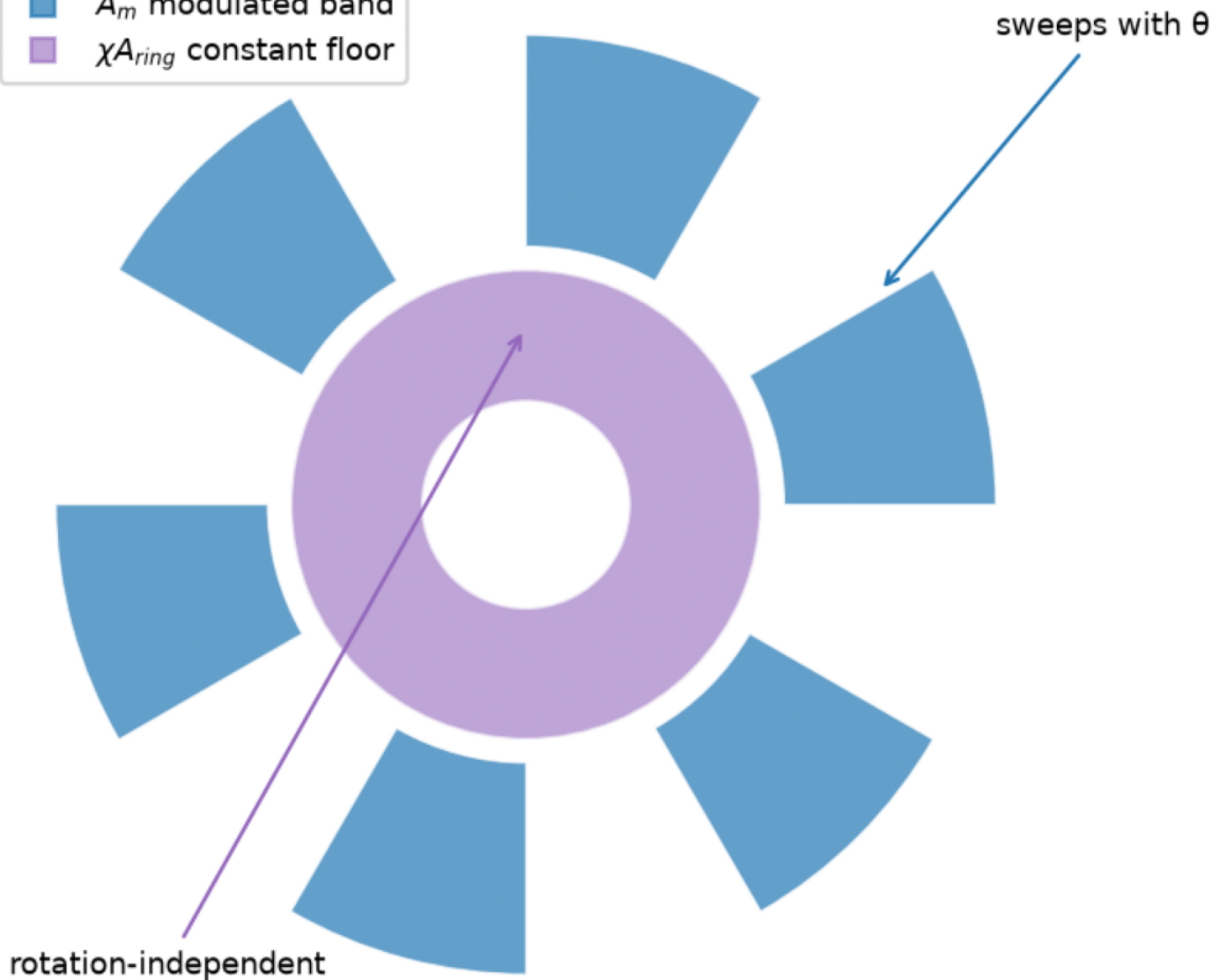
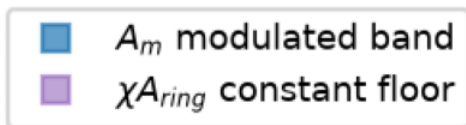
 $\theta = 0$ aligned $\theta = \frac{1}{2}s_\theta$  $\theta = s_\theta$ over gapstriangle, period $2s_\theta$ 

$$\kappa_C = C_{max}/C_{min} \approx 17.5 \quad (\text{pure area ratio} - \varepsilon_0, \varepsilon_r, g \text{ cancel})$$

E4 — Area decomposition (Eq. 4)

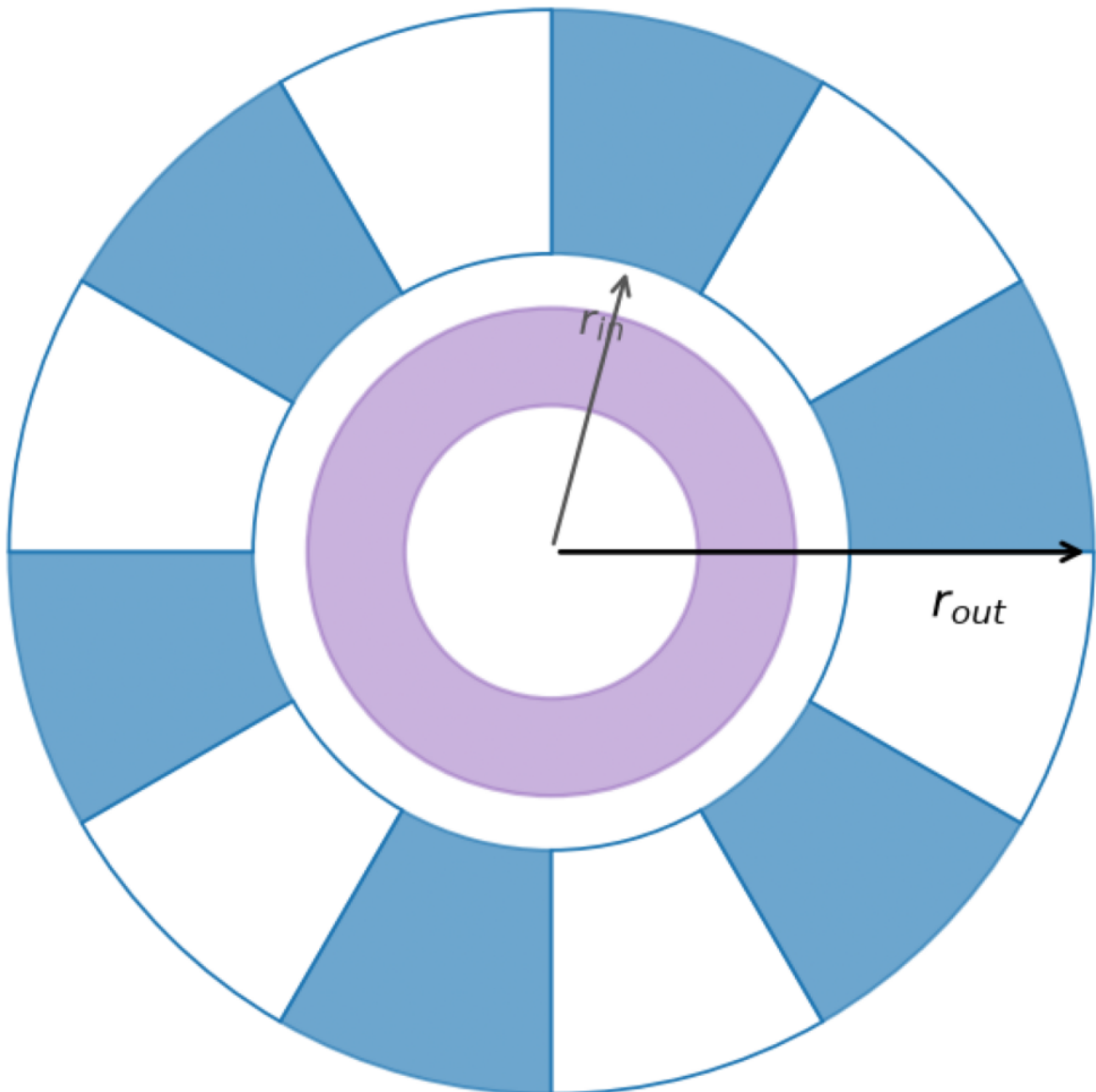
$$A_{ov}(\theta) = A_m f_{ov}(\theta) + \chi A_{ring}$$

(parallel areas add)



E8 — Sector & ring geometry (Eqs. 8, 9)

$$N_{sec} = 12, n_{kept} = 6 \text{ (alternating)} \Rightarrow A_m = \frac{1}{2}\pi(r_{out}^2 - r_{in}^2)$$

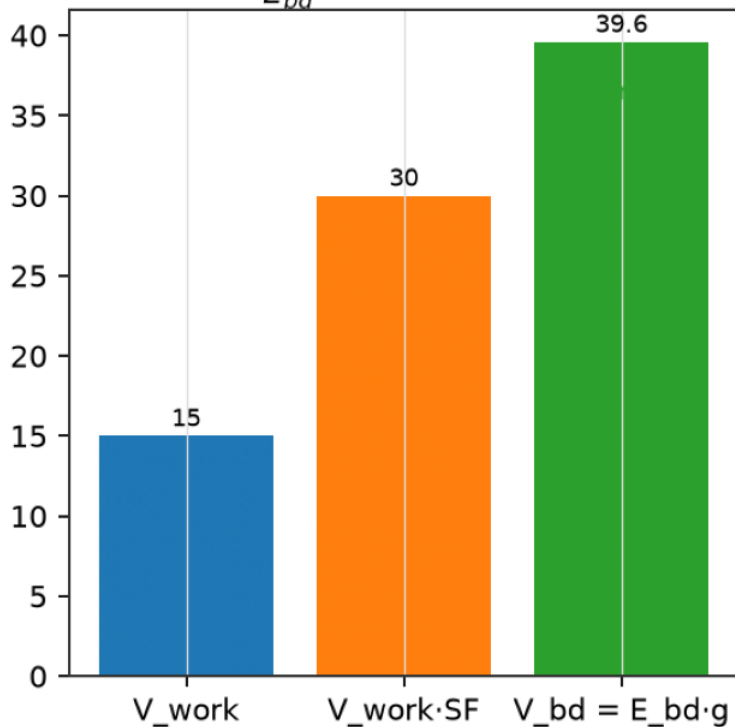
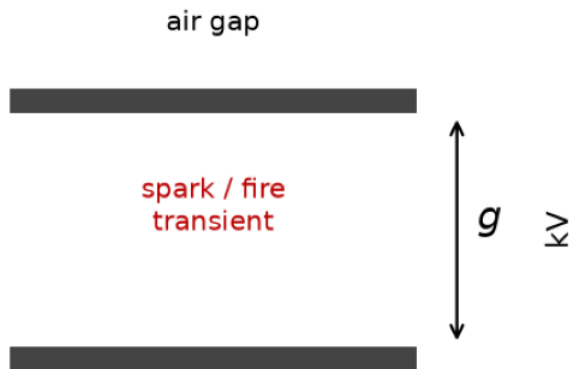


$$\text{ring: } A_{ring} = \pi(r_{ring, out}^2 - r_{ring, in}^2)$$

E10 — Insulate-first gap (Eq. 10)

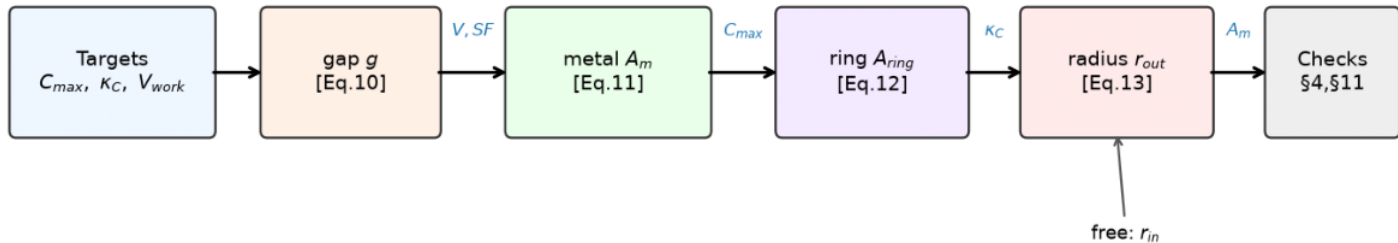
gap = a breakdown decision

$$g = \frac{V_{work}}{E_{bd}} \quad (\text{insulate first})$$



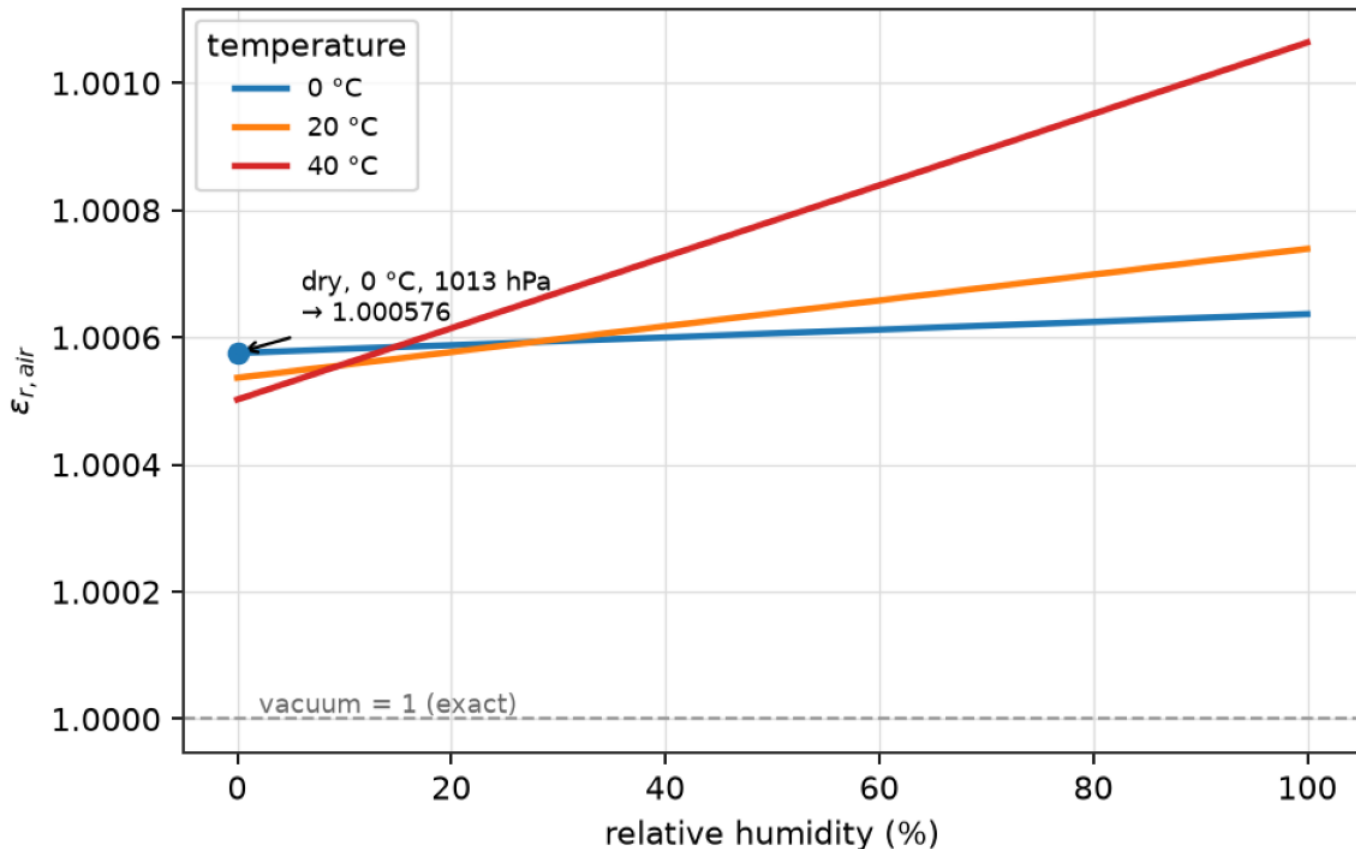
E11 — Inverse-design flow (Eqs. 11, 12, 13)

fixed order — insulate-first: gap before area; κ_C touches only the ring



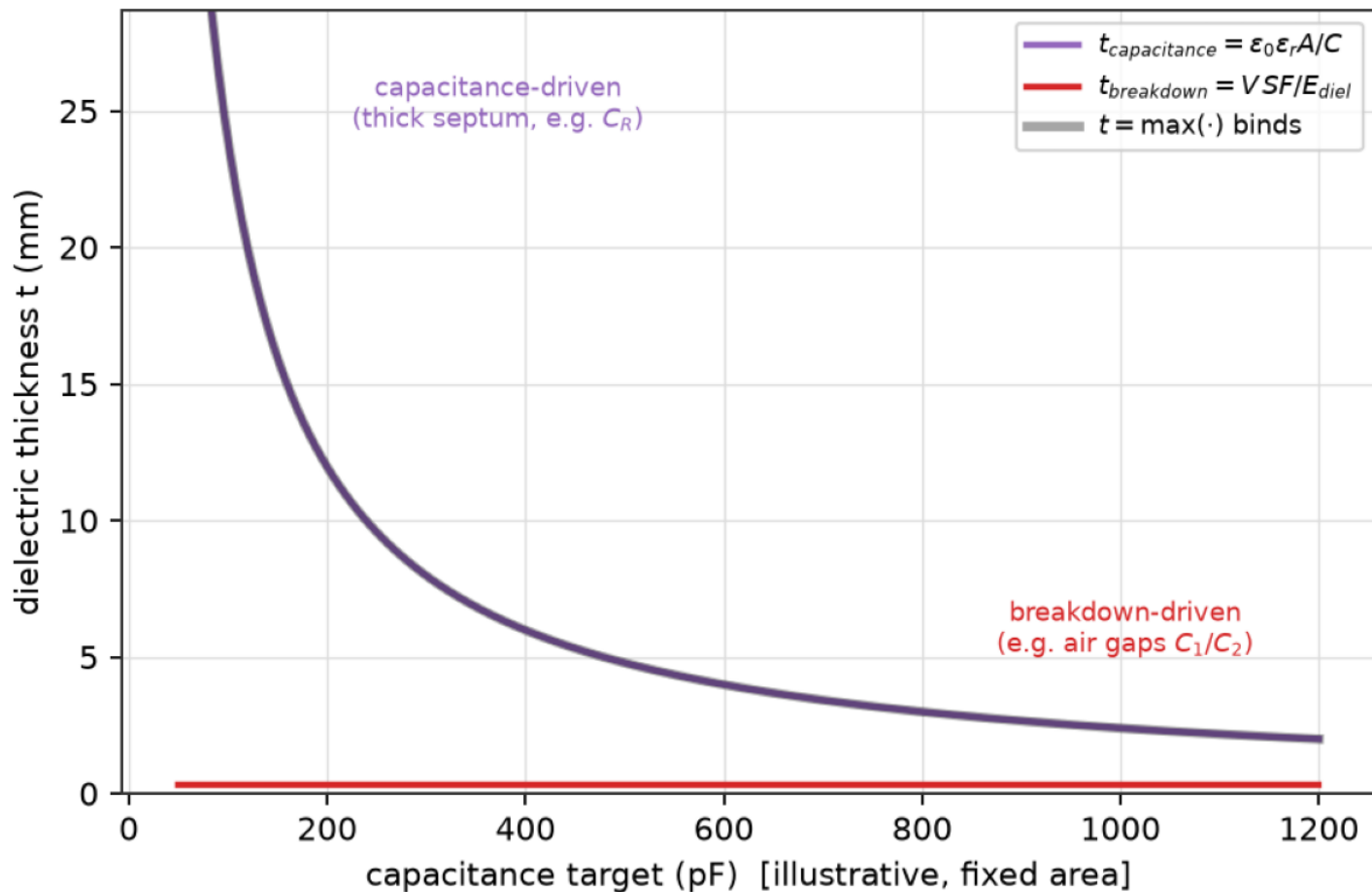
E14 — Moist-air permittivity (Eqs. 14, 15)

$\epsilon_{r,air} = 1 + 2N \times 10^{-6}$ (Smith-Weintraub + Buck) — varies only at the 4th decimal



E16 — Thickness regimes (Eqs. 16, 17, 18)

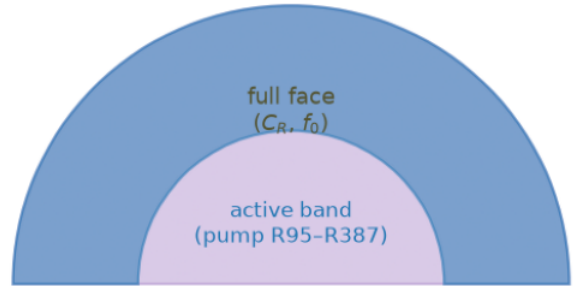
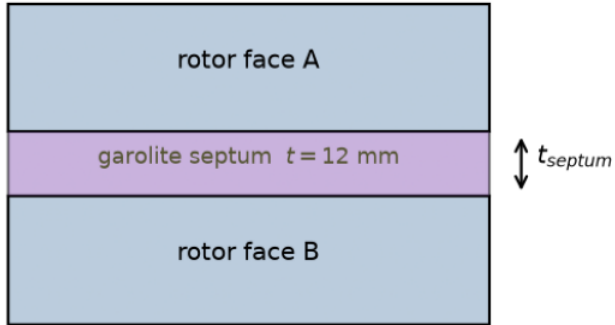
Two independent floors — take the larger (Eq. 18)



E19 — Tank capacitor across the septum (Eq. 19)

$$C_R = \epsilon_0 \epsilon_{r, \text{garolite}} A_{\text{full}} / t_{\text{septum}} = 789 \text{ pF}$$

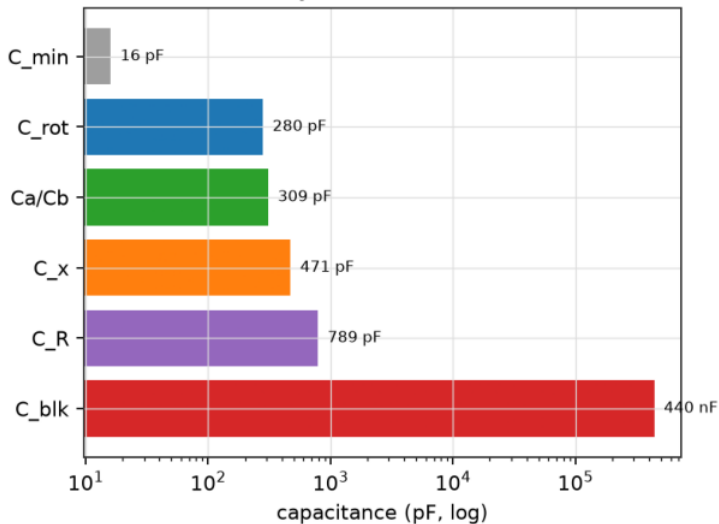
full face $\neq$ squeezed pump band



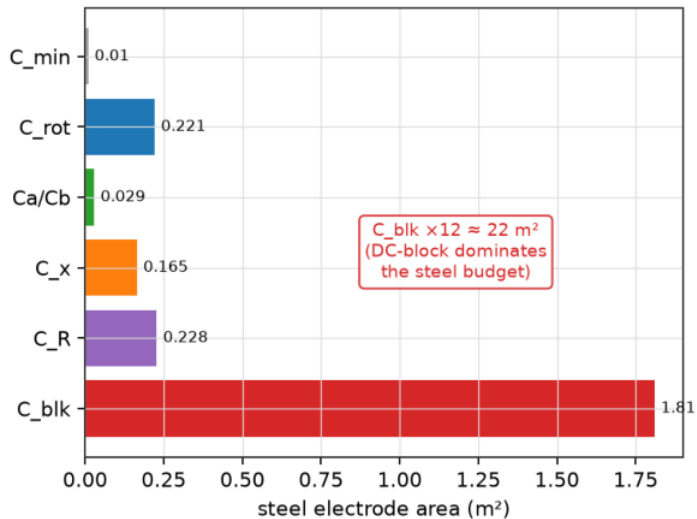
E20_two_ladders.png

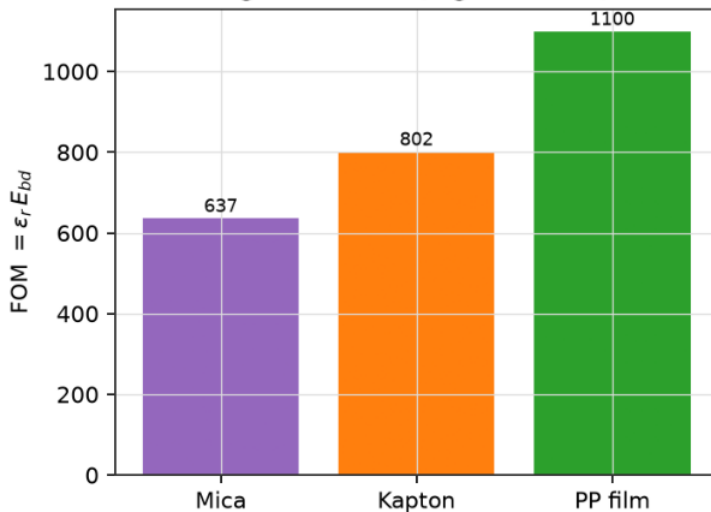
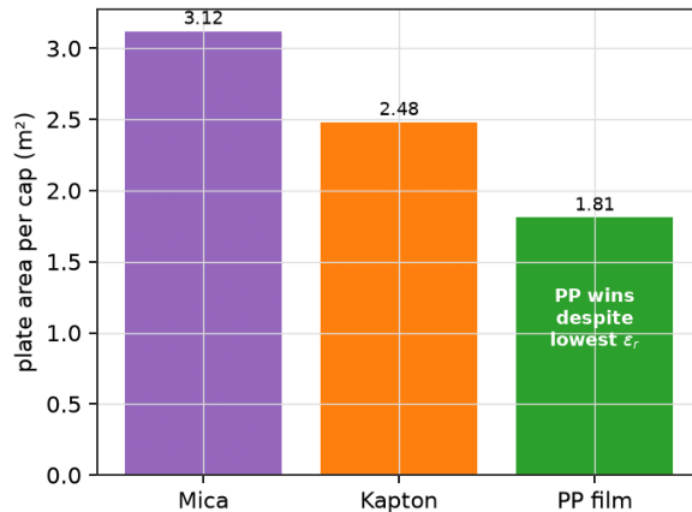
E20 — Two ladders: capacitance vs steel area (Eqs. 20, 21)

capacitance ladder



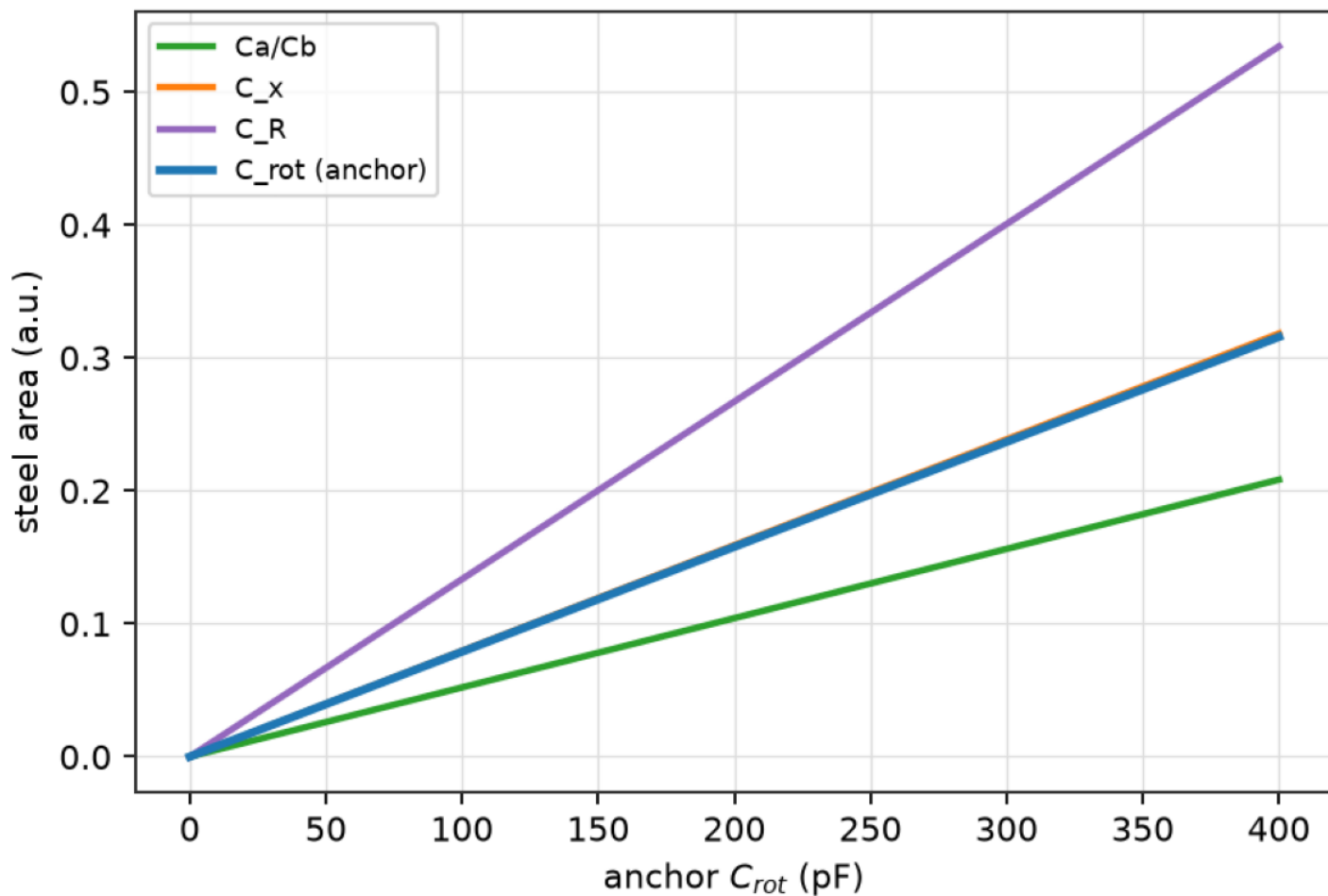
steel-area ladder



E22 — Dielectric FOM and resulting plate area (Eqs. 22, 23, 24)**figure of merit (higher better)****area @ 440 nF/20 kV (lower better)**

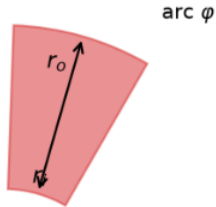
E25 — Linear family scaling (Eq. 25)

fixed (g, ε_r) : $A \propto C$ — pick any anchor, family follows



E26 — Boomerang block cap (Eqs. 26-29) [annotate over real CAD]

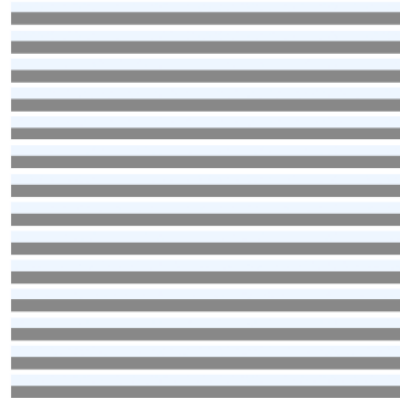
plan: one sector plate $A_{boom} = \frac{1}{2}\phi(r_o^2 - r_i^2)$



section: axial stack

foil / Mylar stack

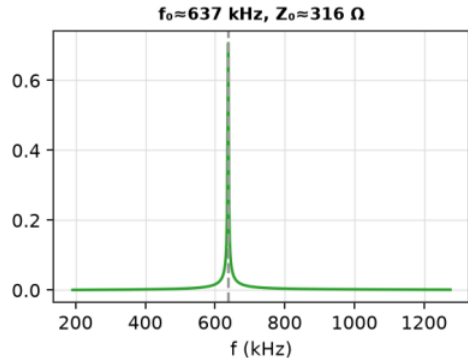
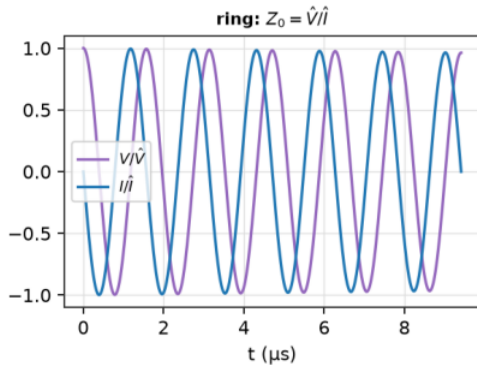
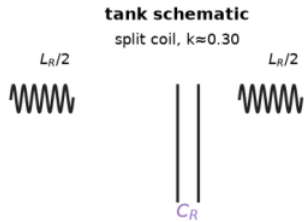
$$N = A_{total}/A_{boom}$$

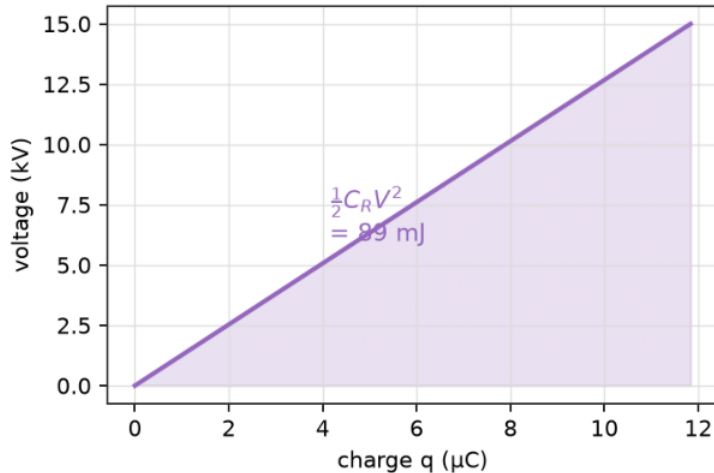
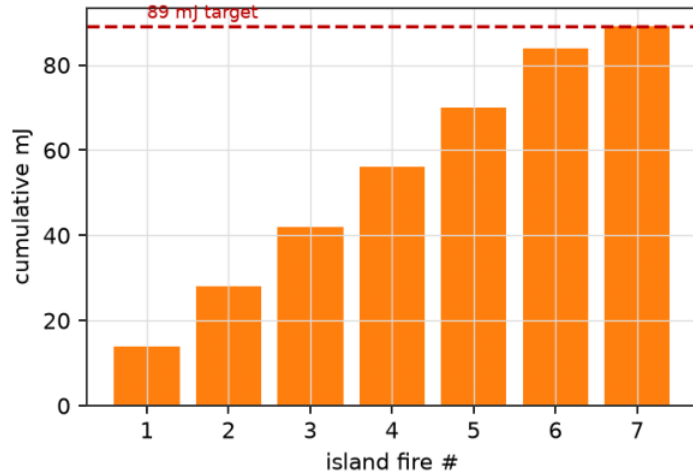


$$H = N(t_{film} + t_{foil})$$

E30_lc_tank.png

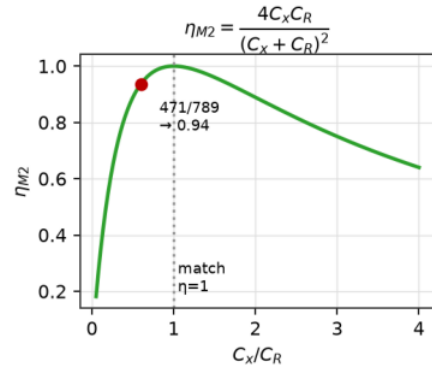
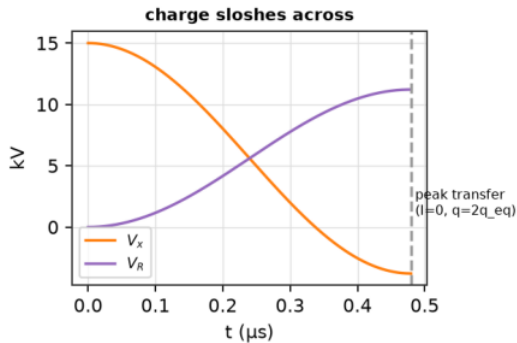
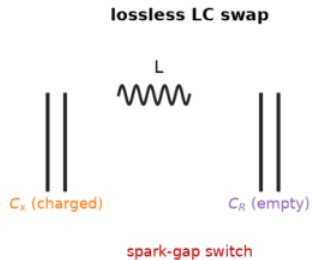
E30 — LC tank: resonance & characteristic impedance (Eqs. 30, 31)



E32 — Reach: $\frac{1}{2} C_R V^2$ and the multi-fire build (Eq. 32)**stored tank energy = triangle area****reached over ~6-7 fires (~14 mJ each)**

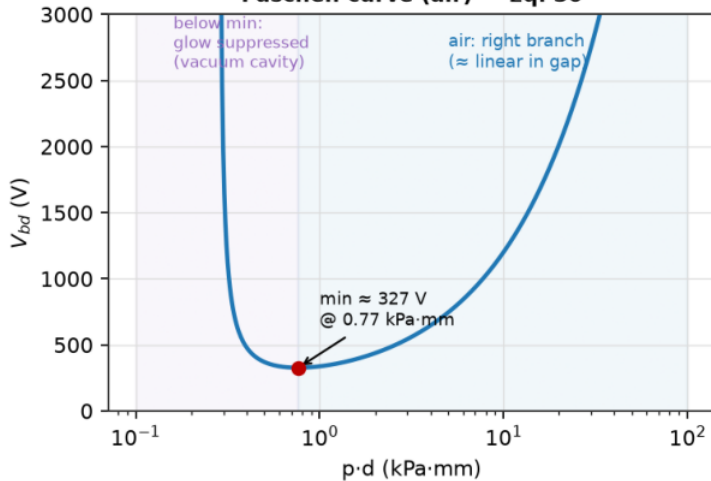
E33_transfer_efficiency.png

E33 — Island→tank transfer efficiency (Eq. 33)

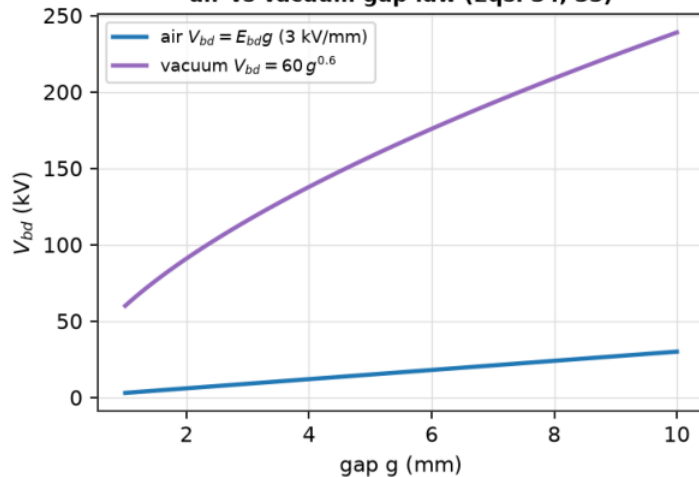


E34 — Breakdown laws & operating regimes (Eqs. 34, 35, 36)

Paschen curve (air) — Eq. 36

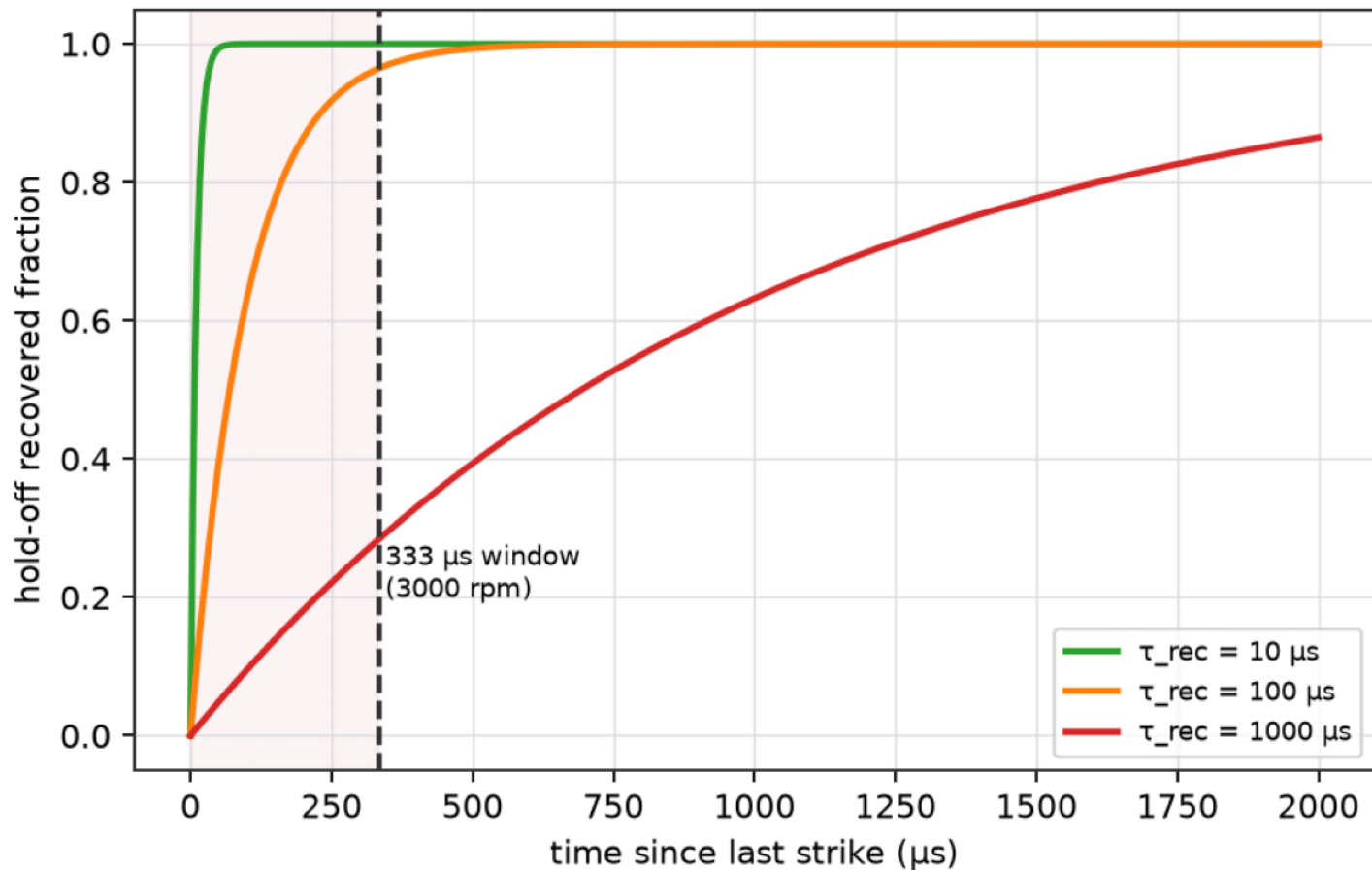


air vs vacuum gap law (Eqs. 34, 35)



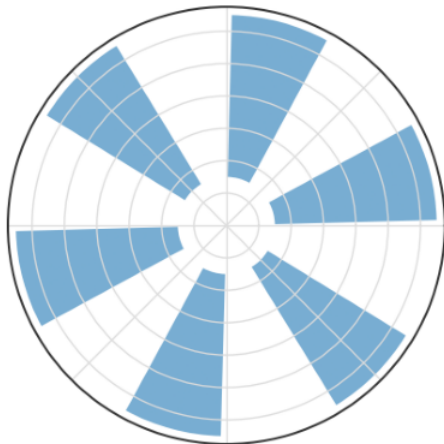
E37 — Hold-off recovery (Eq. 37)

$1 - e^{-t/\tau_{rec}}$ — thermal recovery is the rate limit (Eq. 37)

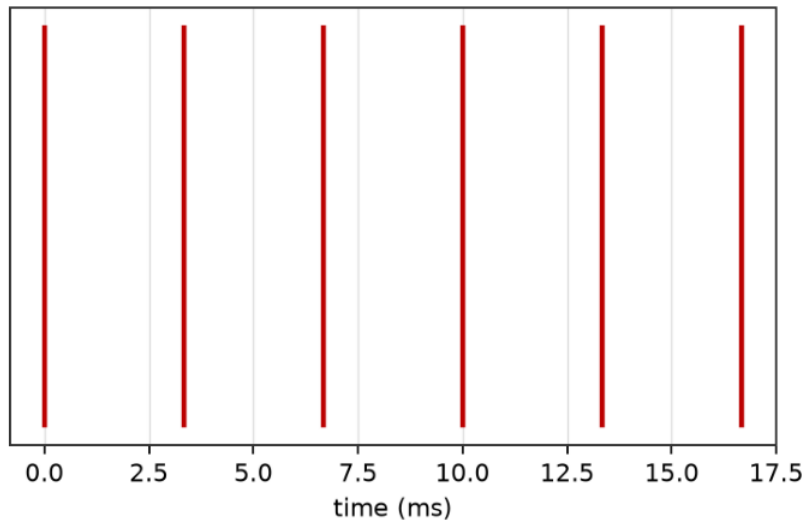


E38 — Firing rate from sector geometry (Eq. 38)

6 kept sectors fire / rev



PRF = $6 \times 3000/60 = 300$ Hz / branch



E39 — Rim speed envelope (Eq. 39)

$$v = 2\pi r_{out} \text{rpm}/60 \quad (r_{out} = 0.49 \text{ m}) \text{ — Eq. 39}$$

